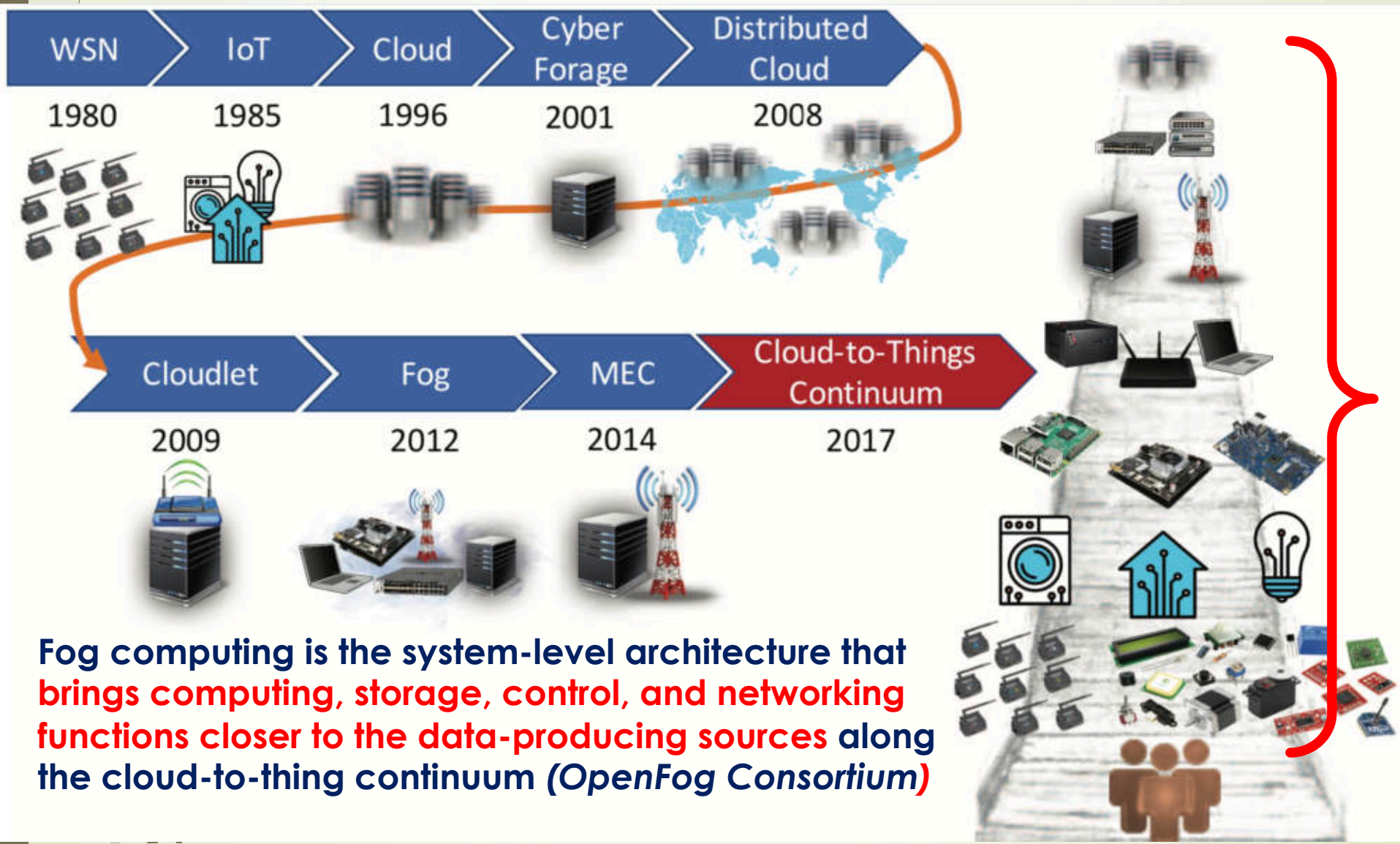


Agenda

- Internet-of-Things (IoT)
- IoT Versus Cyber-Physical Systems (CPS)
- IoT Data Processing
- IoT Analytics

Cloud-to-Things Continuum

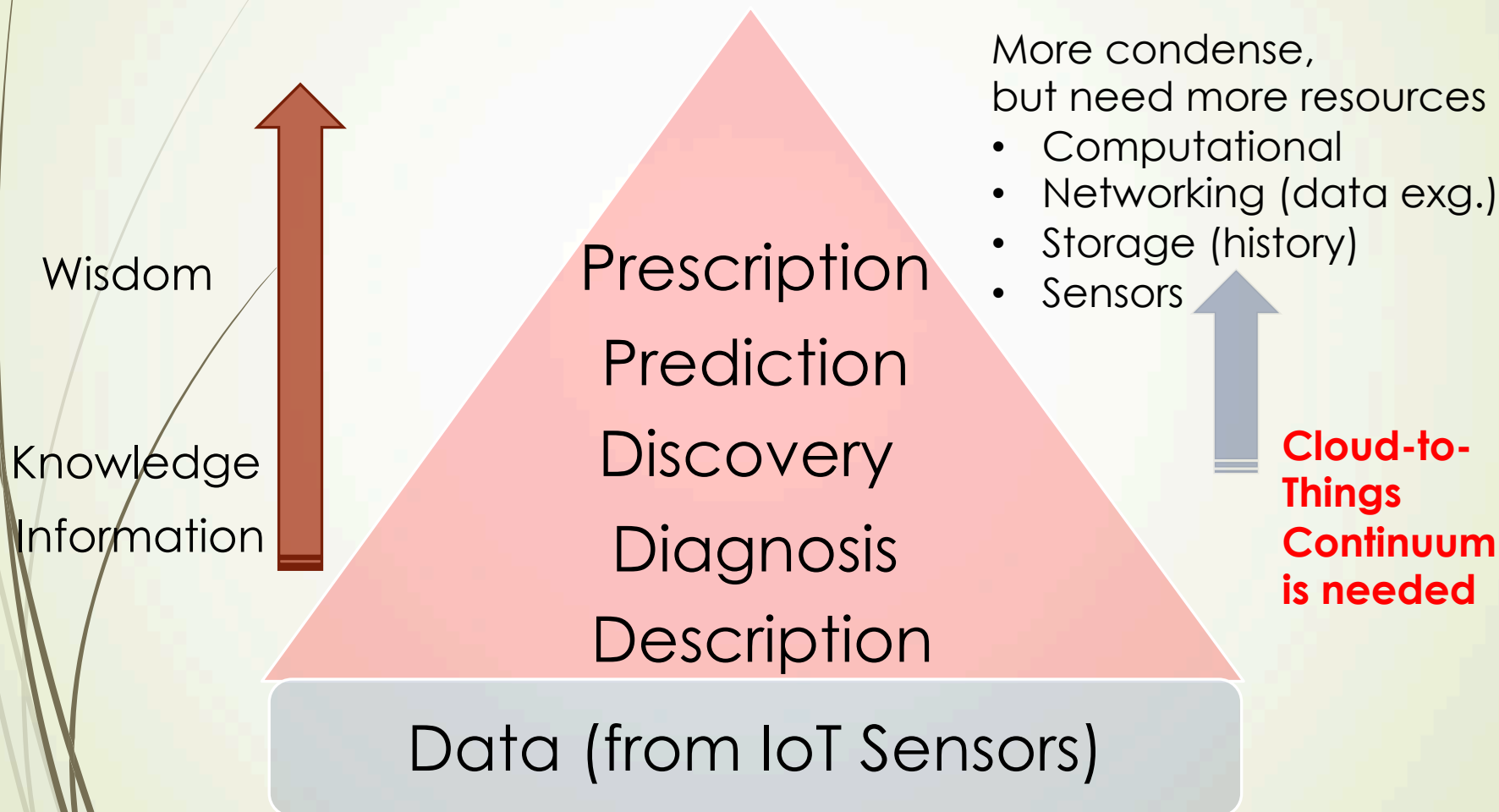
32



What is Analytics

- A process of deriving **knowledge** from data, producing additional values from data [1]
- Examples:
 - Health: wellness recommendations
 - Transport: illegal parking detection
 - Living: building air conditioning
 - Environment: smart energy
 - Industry: smart architecture

What is Knowledge

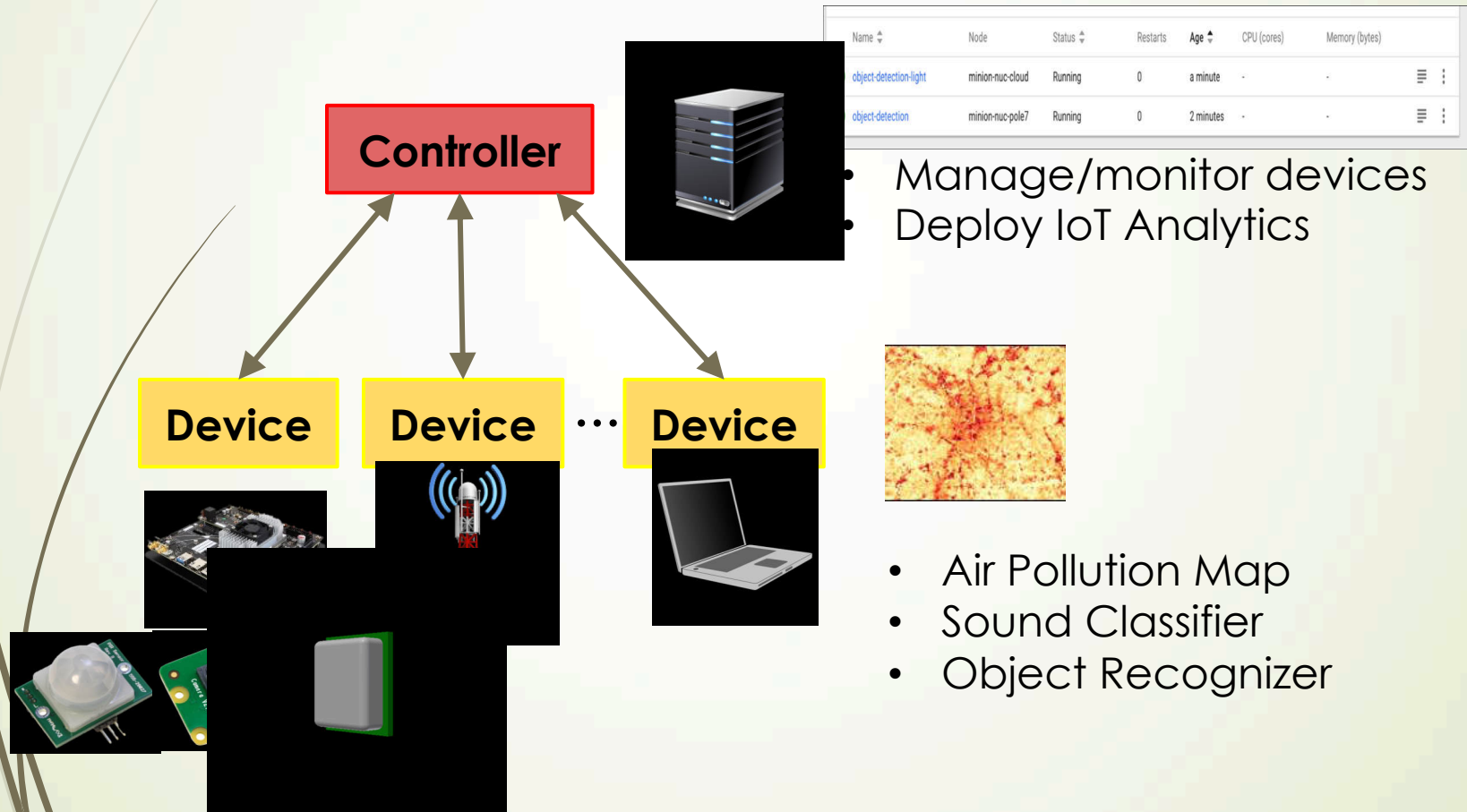


Research Problems for IoT Analytics in Cloud-to-Things Continuum

Analytics Deployment Problem

- **Virtualization/Software-Defined:** multi-tenant supports
- **Quality-of-Service:** quantify/guarantee service quality
- **Resource management:** aggregate resources to meet demand [joint work with Nalini and Yusuf]
- Security and privacy
- Business model

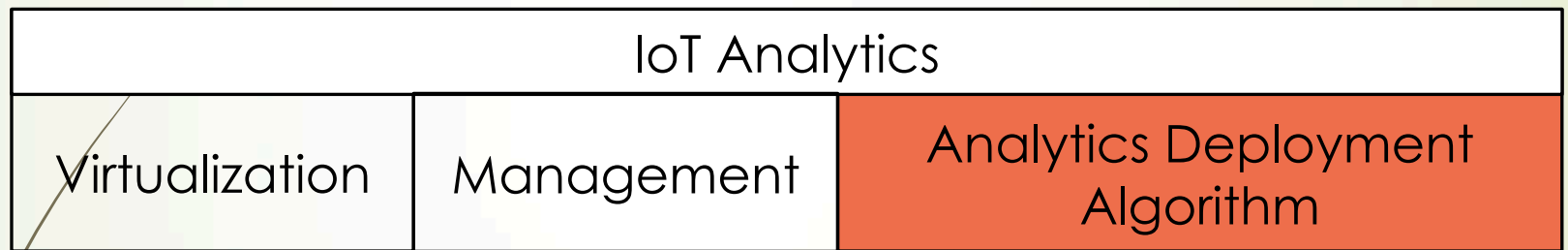
System Topology



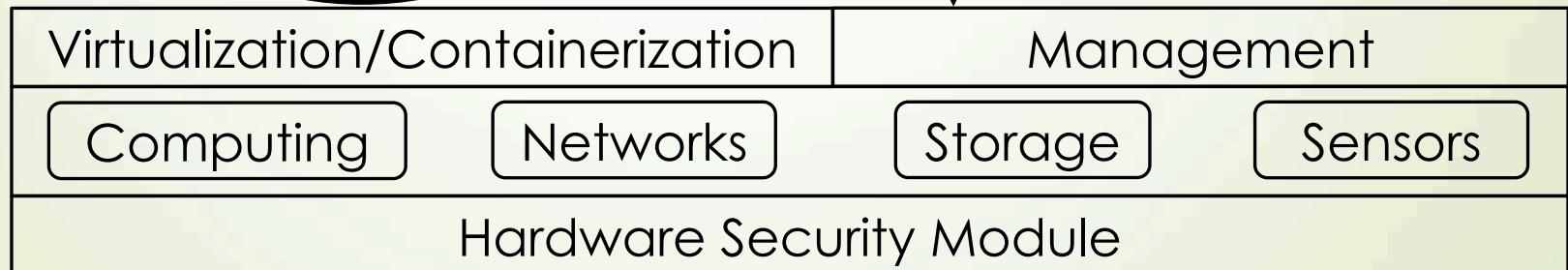
Architecture to Realize Cloud-to-Thing Continuum

Controller

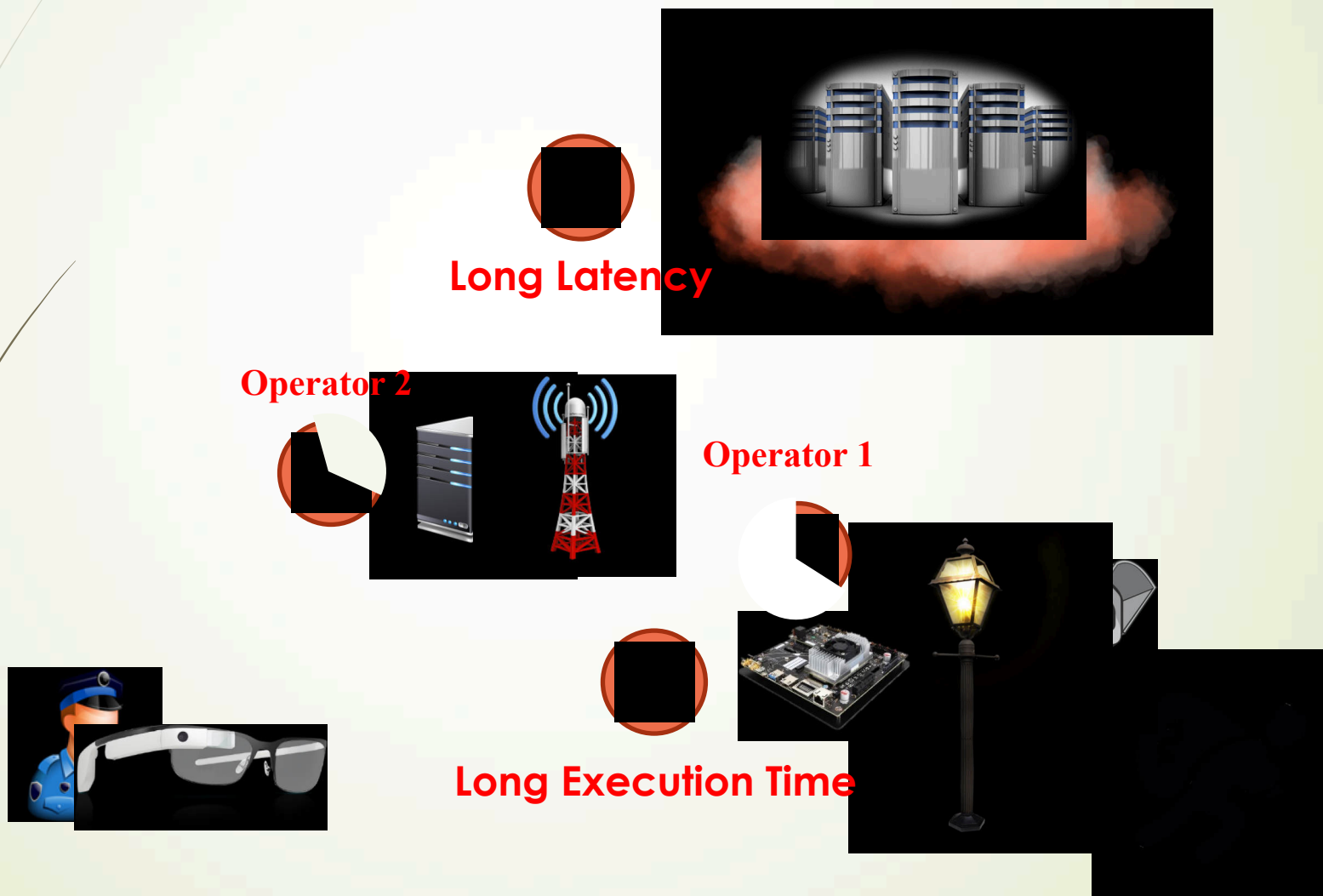
Management



Devices

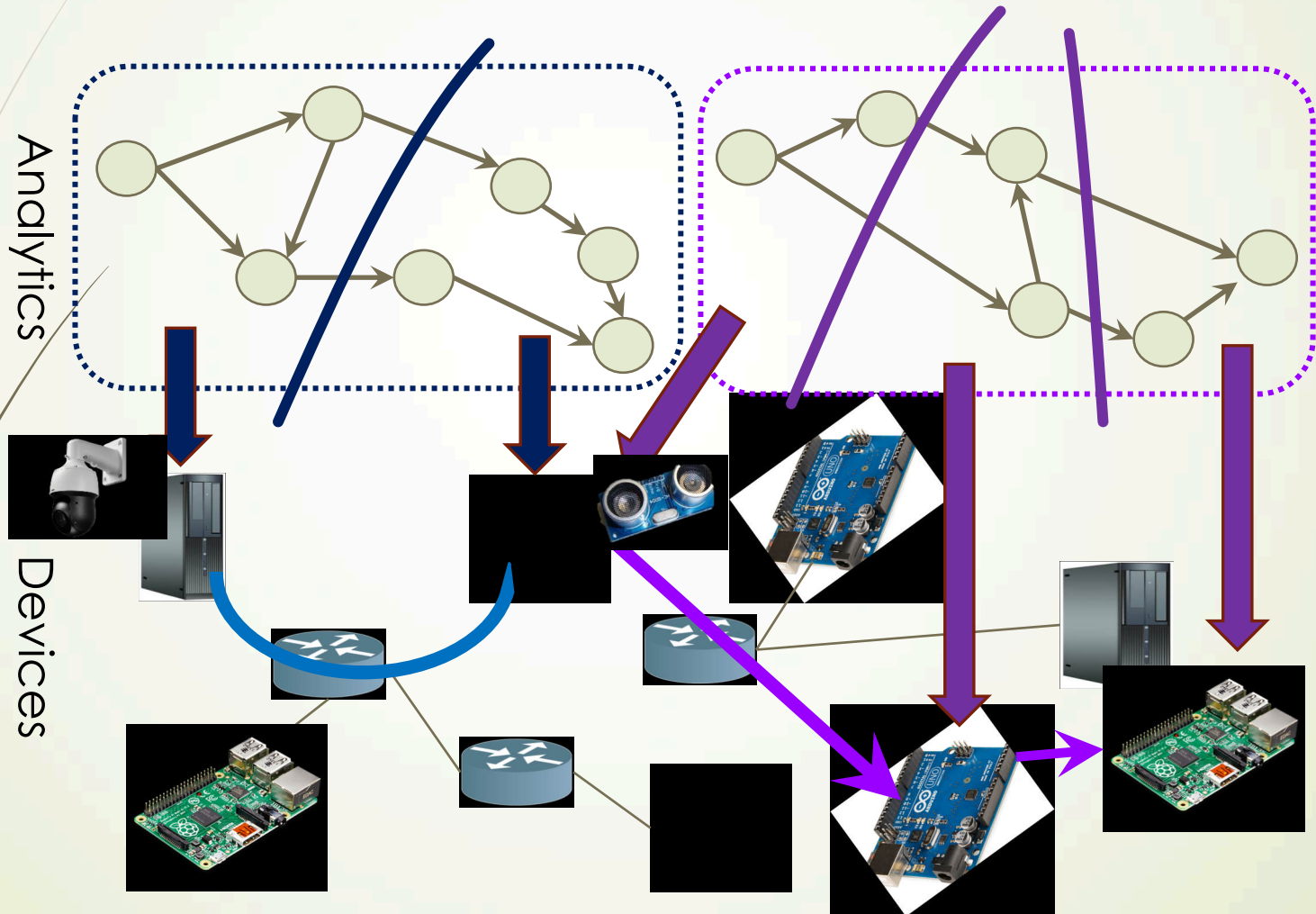


Picking the Right Way to Deploy An IoT Analytics is Hard



Preview: Analytic Deployment Problem

Goal: Maximize number of deployed analytics (requests)



Formal Problem Statement

- ➔ **Problem:** Given a set of requests $\mathbf{Q}$ and a set of fog devices $\mathbf{V}$. Each request consists of: (i) an IoT analytics, (ii) a target QoS, and (iii) a specified location. Devices have $\mathbf{U}$ kinds of resources. Our problem is to determine which request $\mathbf{q} \in \mathbf{Q}$ should be served on which device $\mathbf{v} \in \mathbf{V}$ to maximize number of served requests.

